



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 25: Externalities

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ICS2523 Microeconomics

Lecture 25

Last lecture we studied three of the four models of **oligopoly**

1. **Bertrand** model (price)
2. **Cournot** model (quantity)
3. **Cartel** model (cooperative)
4. **Stackelberg** model
(sequential)

And we finished with oligopolies + repeated games

In today's lecture, we will deepen our knowledge in another market failure:

- **Externalities**

Homework 1 Lecture 25

Stackelberg model

Two identical firms (each owns a spring with a large supply of water). Producers must decide how much to provide the market with the following cost of pumping and bottling q_i liters:

$$C_i(q_i) = cq_i$$

Inverse demand for spring water is: $P(Q) = a - Q$, where $Q = q_1 + q_2$ and $a > c$

Firm 1 is the leader (decides first), and firm 2 is the follower.

Find the price, quantities offered and level of profit for each firm in a Stackelberg model.

Graph the best response functions and identify the points representing the Cournot and Stackelberg equilibria

Homework L25: Solution

Since now firm 1 is the leader and 2 the follower, we need to solve sub-game perfect equilibrium using backward induction:

1. Firm's 2 output choice:

$$q_2 = \frac{a - q_1 - c}{2}$$

(this is the best-response we calculated in Activity 2, for the Cournot model by max firm's 2 profit function with respect to q_2)

2. Firm 1's output choice:

With firm 2's output choice, we fold the game back to solve for firm 1's output choice:

$$\pi_1 = P(Q) \cdot q_1 - c \cdot q_1 = \left[a - q_1 - \left(\frac{a - q_1 - c}{2} \right) - c \right] q_1$$

Homework L25: Solution

$$\pi_1 = P(Q) \cdot q_1 - c \cdot q_1 = \left[a - q_1 - \left(\frac{a - q_1 - c}{2} \right) - c \right] q_1$$

$$\pi_1 = \left[\frac{a}{2} - \frac{q_1}{2} - \frac{c}{2} \right] q_1$$

Maximizing this profit and taking the first-order condition:

$$\frac{\partial \pi_1}{\partial q_1} = \frac{1}{2}(a - q_1 - c - q_1) = 0$$

$$a - 2q_1 - c = 0$$

$$q_1^* = \frac{a - c}{2}$$

Substituting q_1^* into firm 2's best-response function gives:

$$q_2^* = \frac{a - \left(\frac{a - c}{2} \right) - c}{2} = \frac{a - c}{4}$$

Homework L25: Solution

$$\pi_1 = \left[\frac{a}{2} - \frac{q_1^*}{2} - \frac{c}{2} \right] q_1^* = \frac{(a - c)^2}{8}$$

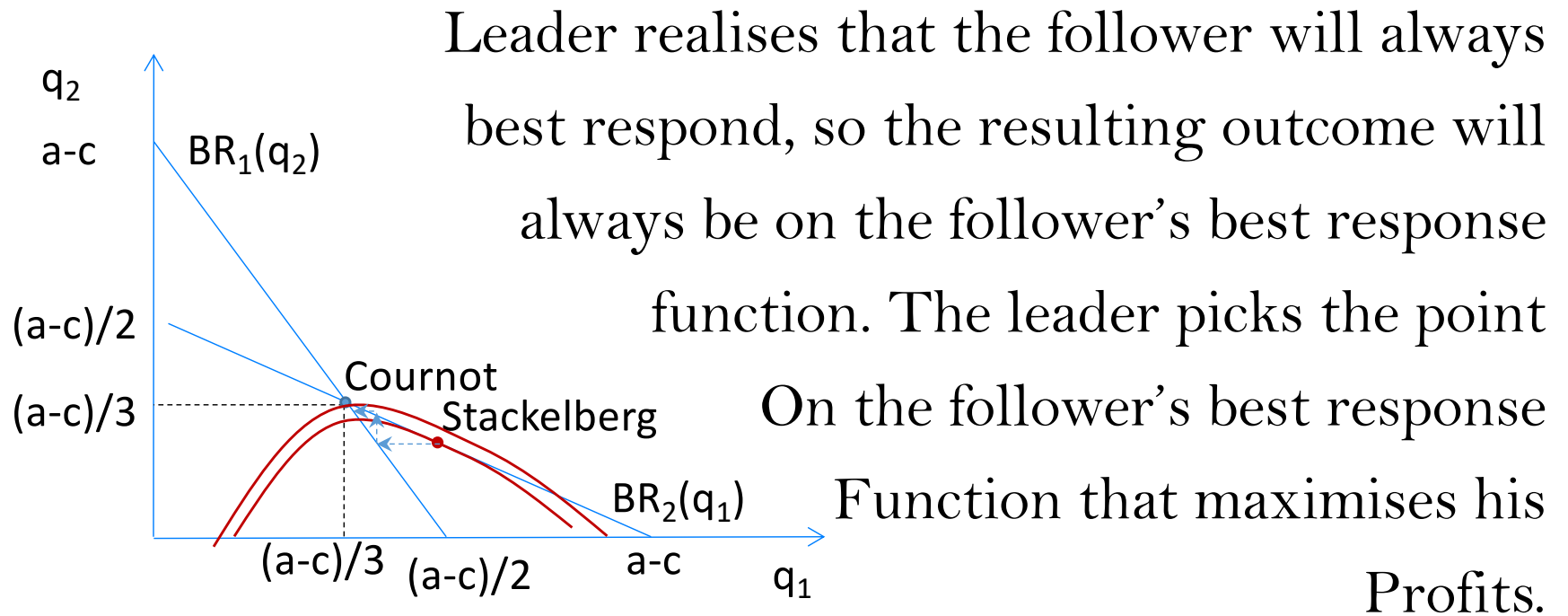
$$\pi_2^* = [a - q_1^* - q_2^* - c] q_2^* = \frac{(a - c)^2}{16}$$

Price is found by replacing the output decisions in the inverse demand function:

$$P(Q) = a - Q = a - q_1^* - q_2^* = a - \frac{a - c}{2} - \frac{a - c}{4}$$
$$P(Q) = \frac{a}{4} + \frac{3c}{4}$$

Homework L25: Solution

Oligopoly: Best-response diagram for Cournot and Stackelberg duopoly



Oligopoly models

Market Failures

Activity: Game

Arrange in groups of 4-5. Each group will be a firm, and you will be competing by price with another group.

		Firm 2	
		Price =1	Price = 10
Firm 1	Price=1	$u_1=10, u_2=10$	$u_1=100, u_2=-50$
	Price=10	$u_1=-50, u_2=100$	$u_1=50, u_2=50$

Externalities

Market Failures

15. Market Failures: Externalities

An **externality** occurs whenever the **activities of one economic actor affect the activities of another** (production function, utility function) in ways that are not reflected in market transactions

This video has very good illustrations of externalities:

<http://www.youtube.com/watch?v=yC5R9WPId0s>

Examples of externalities?

Activity 1

If someone raises the price of some good, is that an externality?

- a) Yes
- b) No



Código de evento
QEDKEX

<https://app.wooclap.com/QEDKEX?from=instruction-slide>

15. Market Failures: Externalities

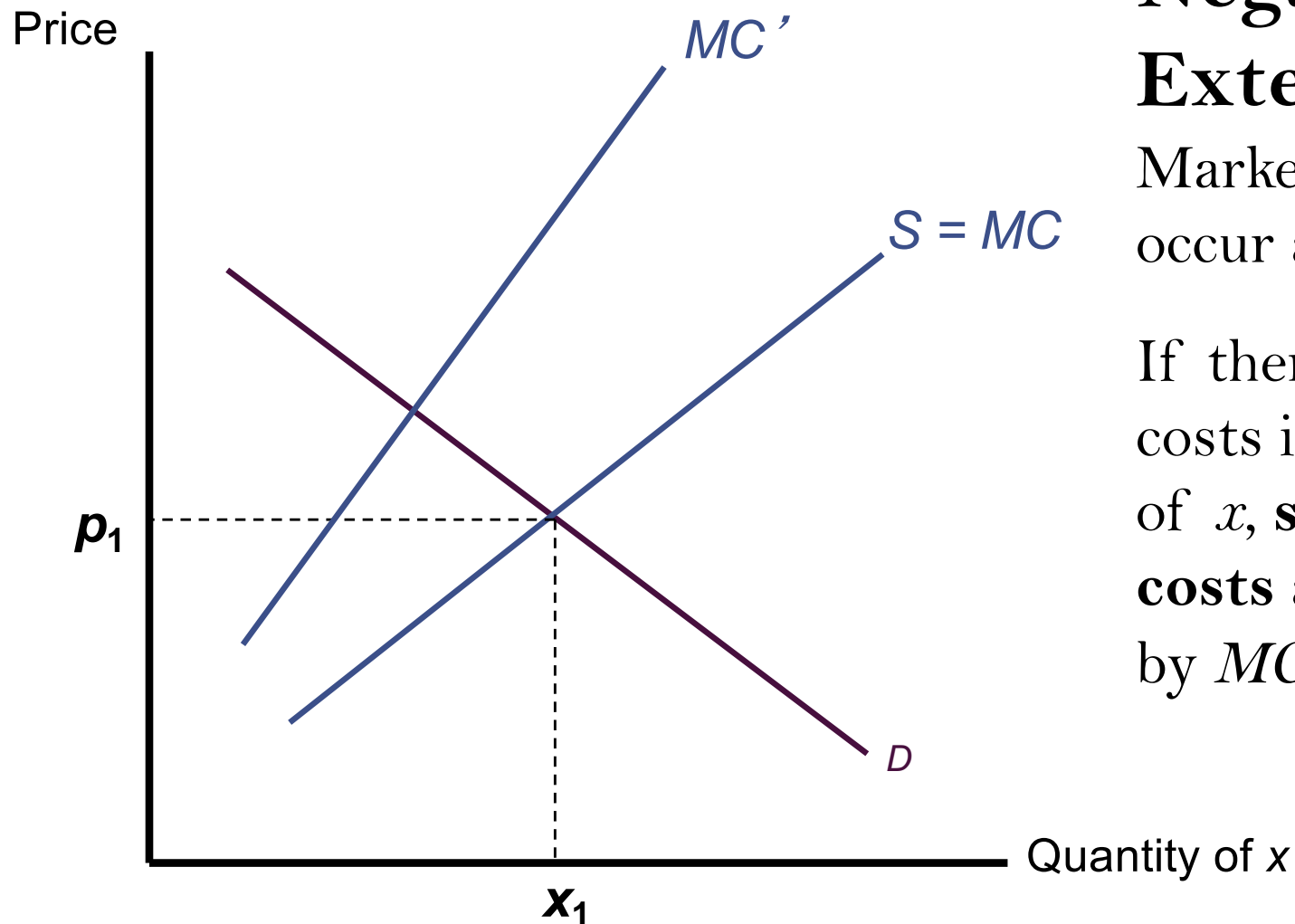
Pecuniary externalities

Effects of one economic agent on another that take place through the market system

Technological externalities

These are the externalities we will refer to in this course because they are the only type with consequences for the efficiency of resource allocation by competitive markets

15. Market Failures: Externalities

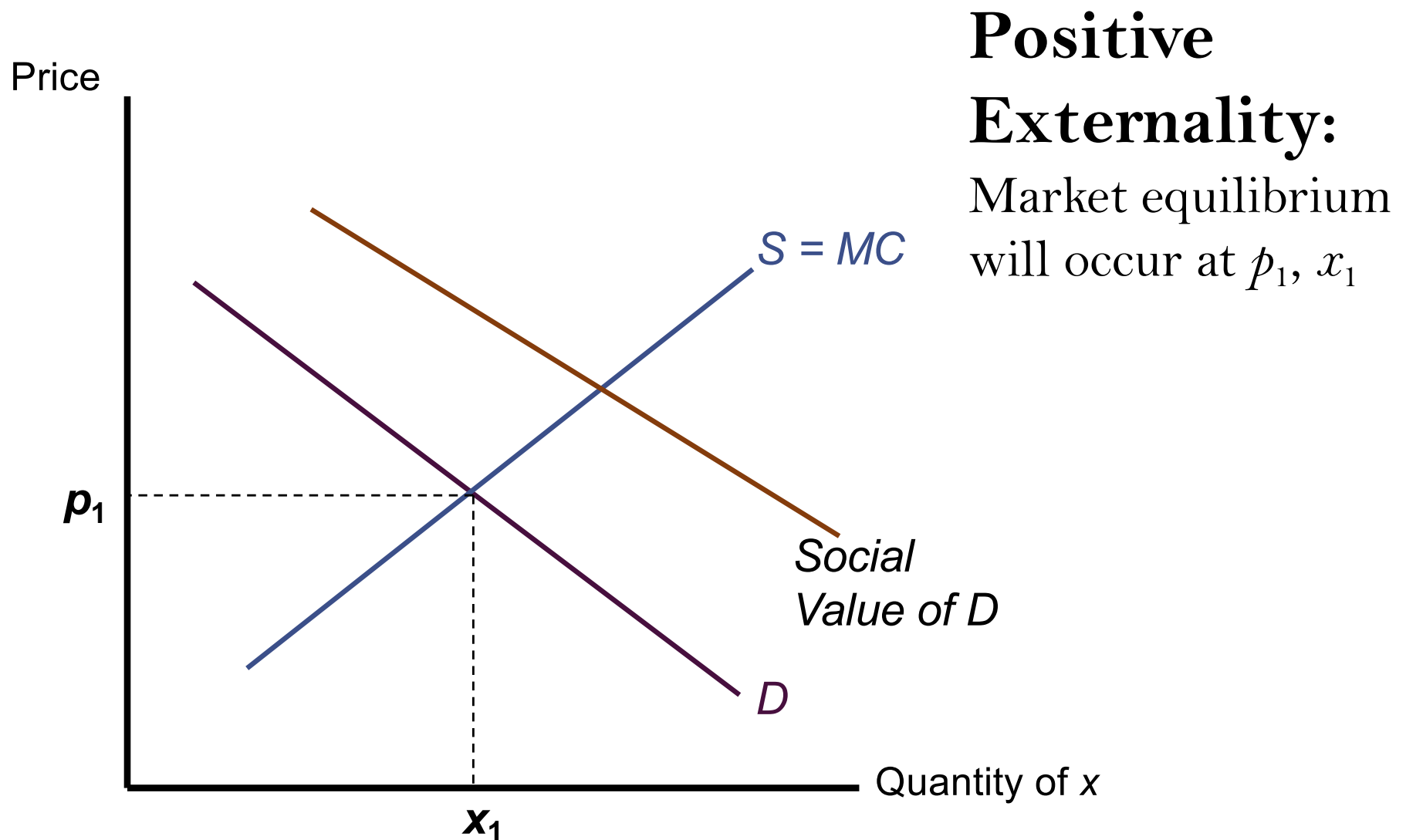


Negative Externality:

Market equilibrium will occur at p_1, x_1

If there are external costs in the production of x , **social marginal costs** are represented by MC'

15. Market Failures: Externalities



15. Market Failures: Externalities

Types of externalities

1. **Interfirm externalities:** two firms, one produces good x and the other good y . The production of good x will have an external effect on the production of y if the output of y depends not only in inputs but also on the level of production of x :

$$y = f(k, l; \mathbf{x})$$

Example: two firms located in a river, with firm y being downstream from x . If x pollutes the river. More $x \rightarrow$ more pollutants $\rightarrow$

$$\partial y / \partial x < 0$$

15. Market Failures: Externalities

Types of externalities

2. **Beneficial externalities:** relationship between two firms might be beneficial $\rightarrow$ production of honey (raising bees) and apples

$$\partial y / \partial x > 0$$

3. **Externalities in utility:** common examples of environmental externalities, where an amount of such activities would enter directly into the individual's utility function $\rightarrow$ utility = $U_S(x_1, x_2, \dots; U_j)$

- If S is altruistic and wants J to be well-off $\rightarrow \partial U_S / \partial U_J > 0$
- If S were envious of J $\rightarrow \partial U_S / \partial U_J < 0$

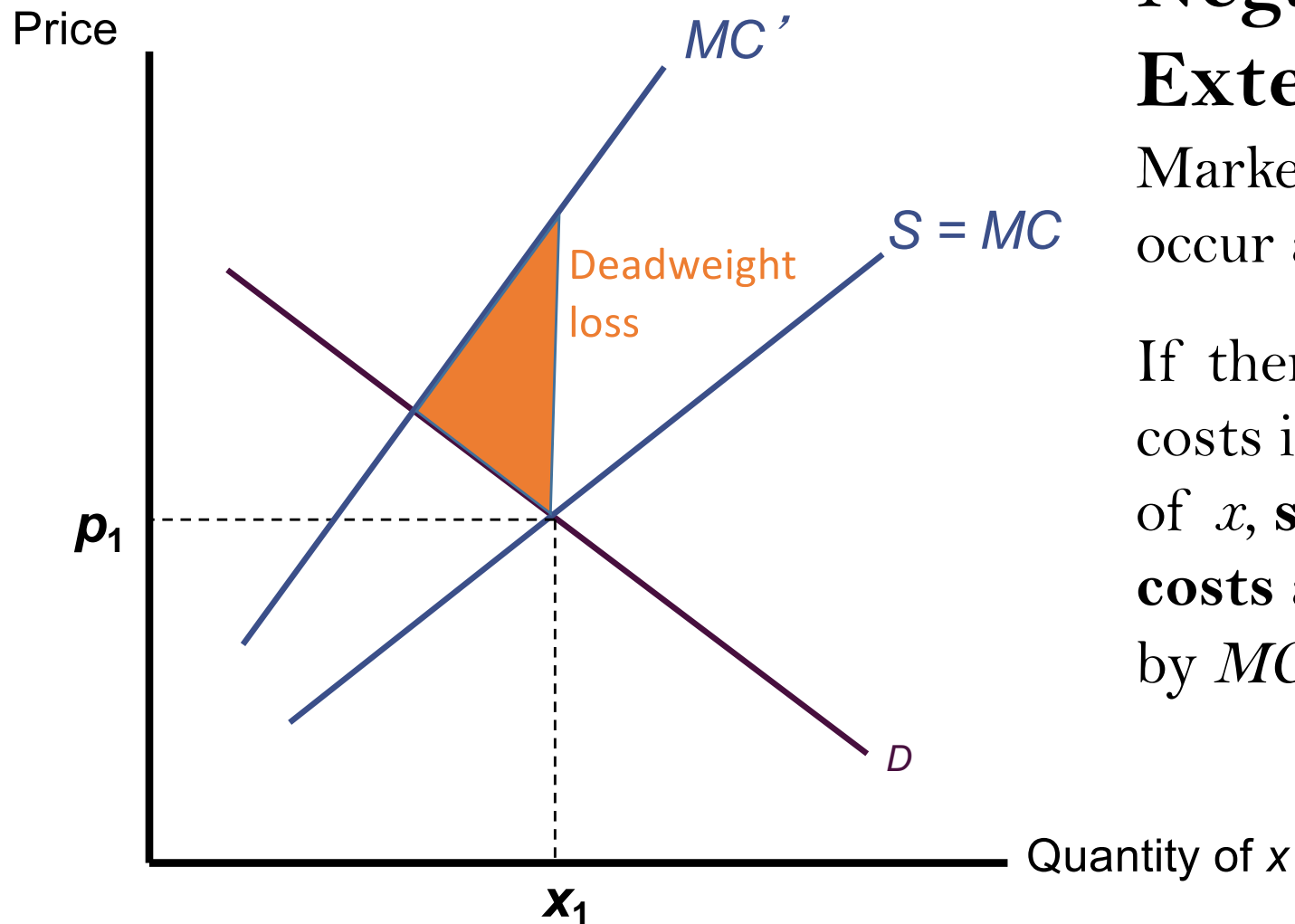
Externalities and Allocative Inefficiency

Market Failures

16. Externalities and Allocative Inefficiency

Externalities lead to **inefficient allocation of resources** because market prices do not accurately reflect the additional costs imposed on or the benefits provided to third parties

15. Market Failures: Externalities

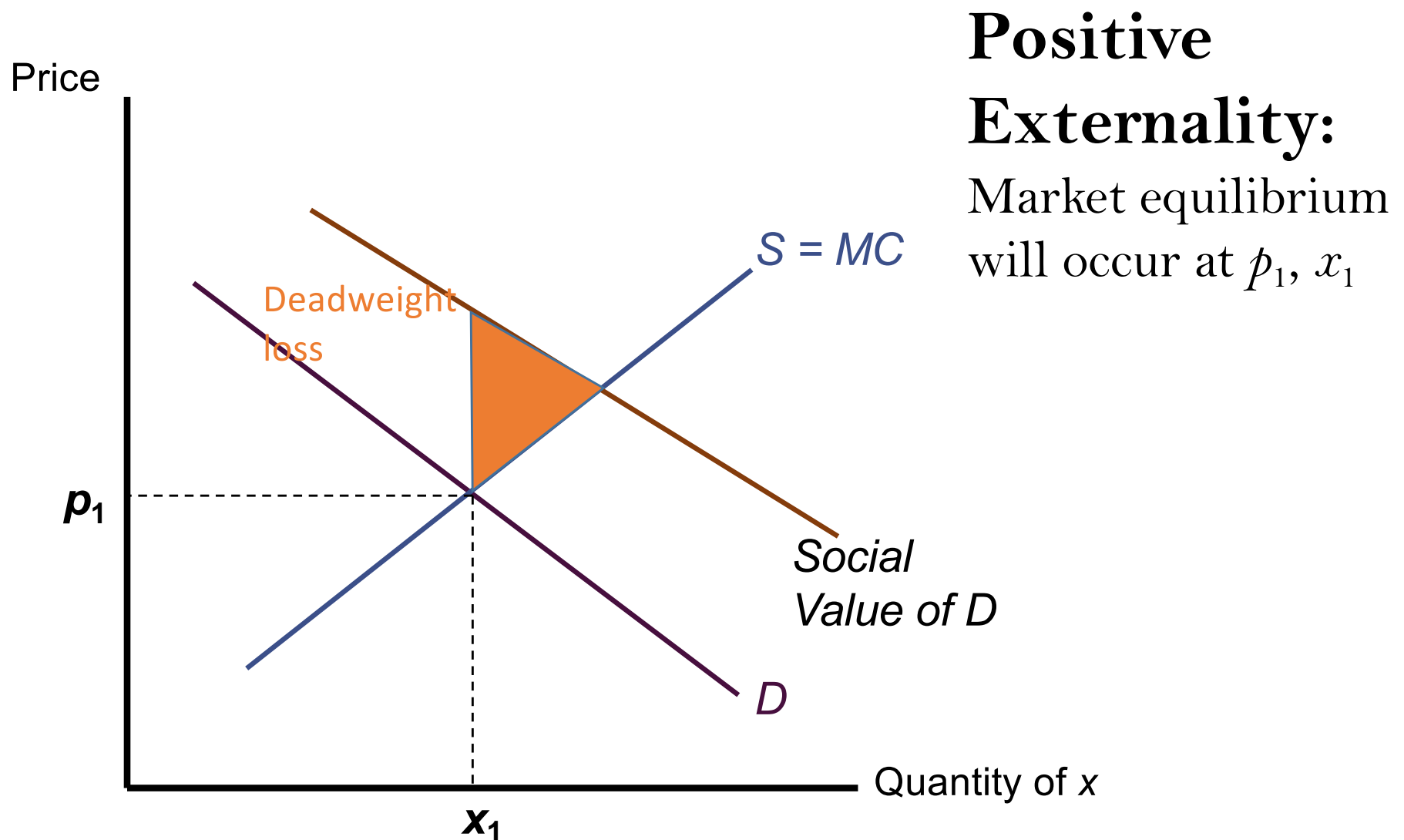


Negative Externality:

Market equilibrium will occur at p_1, x_1

If there are external costs in the production of x , **social marginal costs** are represented by MC'

15. Market Failures: Externalities



16. Externalities and Allocative Inefficiency

An example:

Imagine we have two firms. Firm A produces an output x which it sells in a competitive market. However, producing x imposes a cost ($e(x)$) on firm B.

Letting P be the price of output x , the profits of the two firms are given by:

$$\begin{aligned}\pi_A &= \text{Max}_x P \cdot x - c(x) \\ \pi_B &= -e(x)\end{aligned}$$

The equilibrium amount of output (x_q) is given by $P = c'(x_q)$. However, this output is too large from a social point of view.

Firm A takes account of the **private costs** (costs that it imposes on itself), but it ignores the **social costs** (the private cost plus cost that it imposes on the other firm).

16. Externalities and Allocative Inefficiency

A mathematical example:

In order to determine the efficient amount of output, we ask what will happen if the two firms merged so as to **internalize** the externality. In this case, the merged firm would maximize total profits:

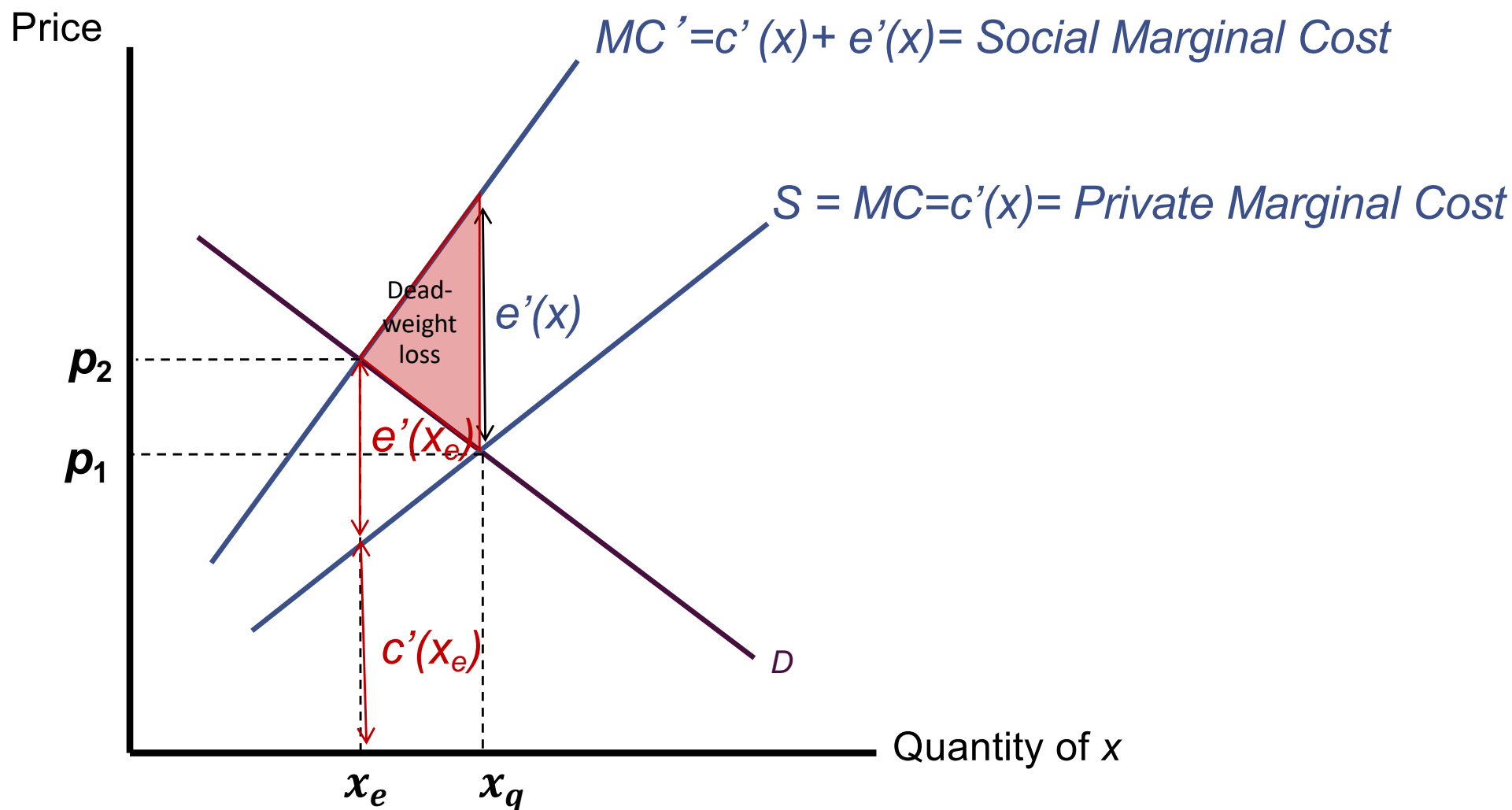
$$\pi = \text{Max}_x P \cdot x - c(x) - e(x)$$

And this problem has first-order condition:

$$P = c'(x_e) + e'(x_e)$$

The output x_e is an efficient amount of output, it is characterized by **price being equal to marginal social cost**.

16. Externalities and Allocative Inefficiency



Graphs taken from: www.4shared.com

16. Externalities and Allocative Inefficiency

A more complex model

- We will assume that we have an economy of one person
- This person's utility depends on quantities of x and y consumed $\rightarrow x_c, y_c$
- Utility = $U(x_c, y_c)$

16. Externalities and Allocative Inefficiency

- This person has initial stocks (endowments) of x and y are x^* and y^* , respectively, and may consume them directly or use them as intermediary goods in production
- We will assume that good x is produced only using good $y \rightarrow x_o = f(y_i)$
- Output of good y depends not only on how much x is used as an input in the production process, but also on the x production level itself

$\rightarrow y_o = g(x_i, x_o)$, where $\frac{\partial g}{\partial x_i} > 0, \frac{\partial g}{\partial x_o} < 0$

16. Externalities and Allocative Inefficiency

- Quantities of each good are constrained by the initial stocks available and by the additional production that takes place

$$x_c + x_i = x_o + x^*$$

$$y_c + y_i = y_o + y^*$$

Activity 2

Write down the economic model for this example and find the first order conditions.

HINTS:

- What are we maximizing?
- Which are our variables? (there are six)
- Which are the restrictions? (there are four)

16. Externalities and Allocative Inefficiency

Now, with equations (1) and (2):
$$MRS = \frac{\partial U / \partial x_c}{\partial U / \partial y_c} = \frac{\lambda_3}{\lambda_4} \quad (*)$$

With equations (3) and (4):
$$RTS = \frac{\lambda_3}{\lambda_4} = \frac{\lambda_2 \partial g / \partial x_i}{\lambda_1 \partial f / \partial y_i} \quad (**)$$

To achieve efficiency in x production, we must consider the externality that this production poses to y $\rightarrow$ (4), (5) and (6)

$$MRS = \frac{\lambda_3}{\lambda_4} = \frac{\overset{\text{Eq 5}}{-\lambda_1 + \lambda_2 \partial g / \partial x_0}}{\lambda_4} = \frac{-\lambda_1}{\lambda_4} + \frac{\lambda_2 \partial g / \partial x_0}{\lambda_4}$$

$$MRS = \frac{\overset{\text{Eq 4}}{1}}{\partial f / \partial y_i} + \partial g / \partial x_0 \quad \text{Eq 6: } \lambda_2 = \lambda_4$$

(***) $\rightarrow$ Second term represents the negative impact that added x production has on y output

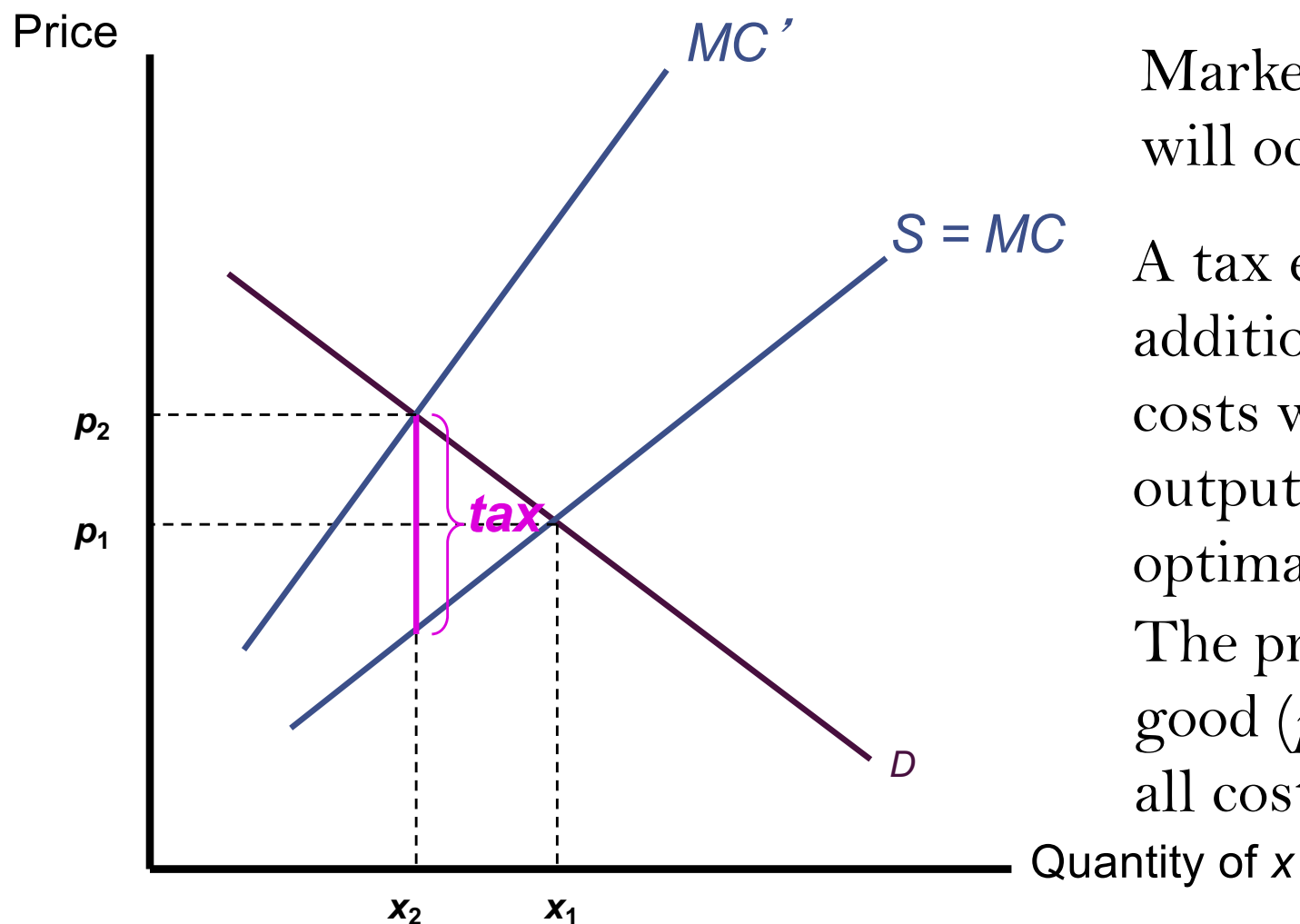
16. Externalities and Allocative Inefficiency

The problem with negative externalities is that output of the externality producing activity is too high under a market-determined equilibrium.

Pigou in 1920 said that the most direct solution would be to tax the externality-creating entity (and he also recognized the importance of subsidizing goods that yield positive externalities)

→ **Pigovian tax**

16. Externalities and Allocative Inefficiency



Market equilibrium will occur at p_1, x_1

A tax equal to these additional marginal costs will reduce output to the socially optimal level (x_2)

The price paid for the good (p_2) now reflects all costs

Homework Lecture 25

Suppose two newsprint producers are located along a river. The upstream firm has a production function of the form:

$$x = f(l_x) = 2000\sqrt{l_x}$$

Where l_x is the number of workers hired per day and x is newsprint output in feet. The downstream firm has a similar production function, but its output may be affected by the chemicals the upstream firm dumps into the river as it produces more x :

$$y = f(l_y, x) = 2000\sqrt{l_y} \cdot (1 + \alpha x)$$

If $\alpha = 0$, then the upstream firm's production has no effect on the downstream firm, but if $\alpha < 0$, then an increase in x harms the downstream firm's production. The newsprint sells for $P = \$1$ per foot and workers earn $w = \$100$ per day.

- If $\alpha = 0$, find how many workers each firm hires and how much newsprint will be produced by each firm.
- If $\alpha = -1/40000$, find how many workers each firm hires and how much newsprint will be produced by each firm.
- If both firms merge, find how many workers each firm hires and how much newsprint will be produced by each firm

Solutions to the externalities problem

Market Failures

17. Solutions to the externalities problem

1. Merge
2. Government provision
3. Subsidies / grants (positive externalities)
4. Taxation
5. Emissions standards and Fees
6. Pollution rights (Coase Theorem)

17. Solutions to the externalities problem

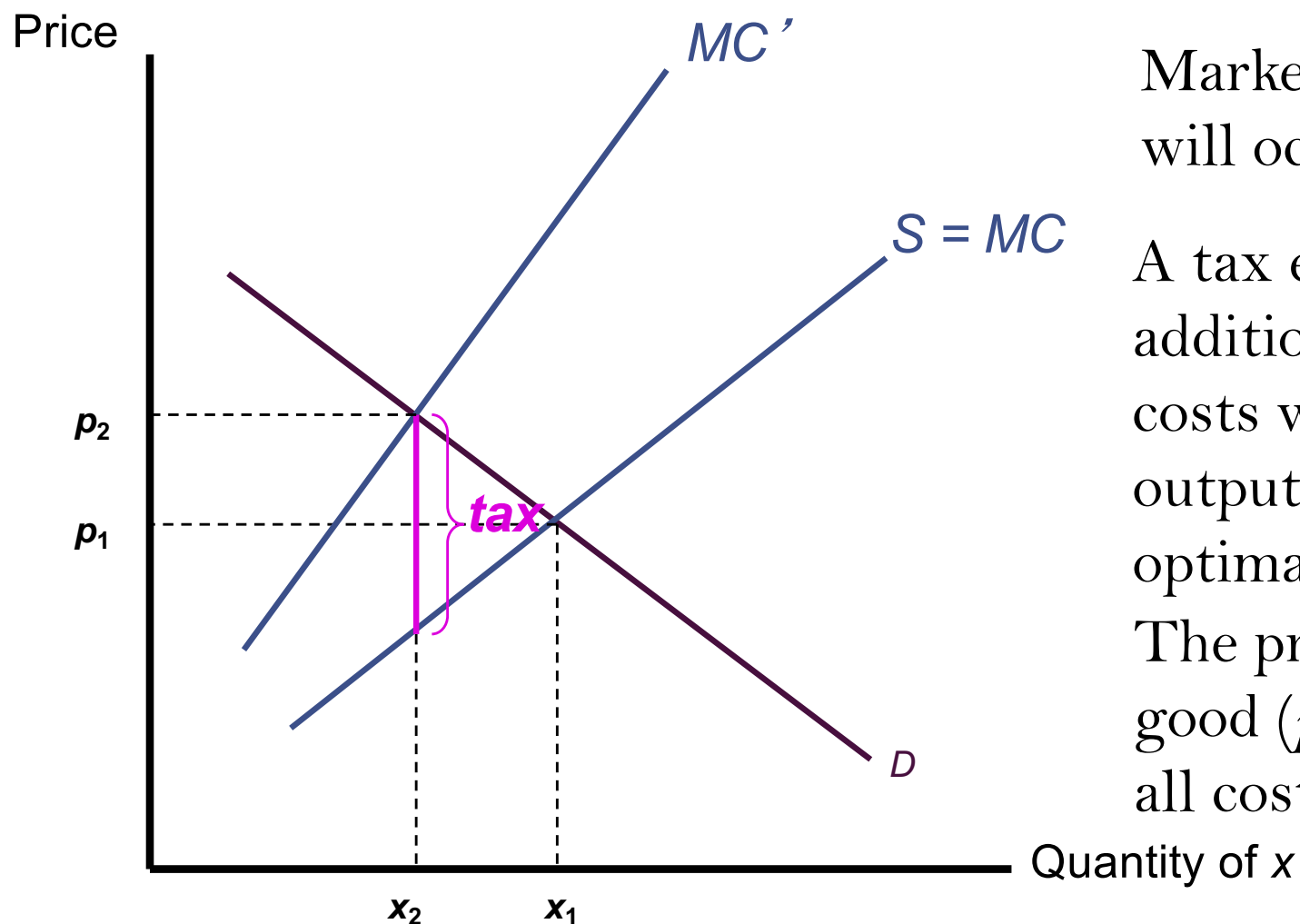
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17. Solutions to the externalities problem: **Taxation**



Market equilibrium will occur at p_1, x_1

A tax equal to these additional marginal costs will reduce output to the socially optimal level (x_2)

The price paid for the good (p_2) now reflects all costs

17. Solutions to the externalities problem: **Taxation**

Taxation $\rightarrow$ the optimal Pigovian tax is set to

$$t = -P_y \partial g / \partial x_o$$

(Remember that production of y depends on how much x is used in its production (x_i) and the total quantity of x produced (x_o)) $\rightarrow y_o = g(x_i, x_o)$,

Per unit tax on good x should reflect the **marginal harm** that x does in reducing y output, valued at the market price of good y .

With the optimal tax, firm x now faces a net price of $(p_x - t)$ and will choose y input according to

$$p_y = (p_x - t) \partial f / \partial y$$

17. Solutions to the externalities problem: **Taxation**

The resulting allocation of resources will achieve

$$MRS = p_x/p_y = (1/\partial f/\partial y) + t/p_y = (1/\partial f/\partial y) - \partial g/\partial x_o$$

Problem → depends on the **output level** (should be different for different output levels)

→ regulators must know firm y's production function

17. Solutions to the externality problem:

Emissions Standards and Fees

Emissions Standard

- Establish legal limit to the level/quantity of emissions each firm may produce

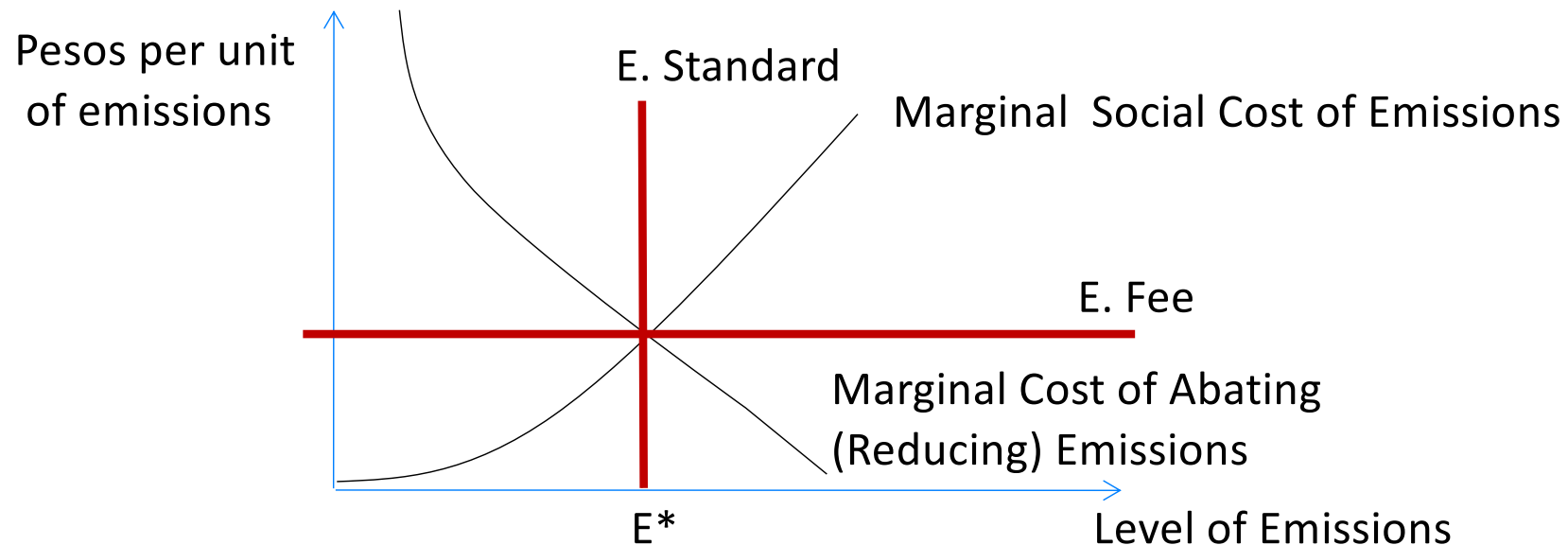
Emissions Fee

- Each firm pays for each kg of emissions produced

17. Solutions to the externality problem:

Emissions Standards and Fees

Which one is better?



Activity 3

Which one is better: **emissions standard** or **emissions fee**? Why?

17. Solutions to the externality problem: **Pollution rights**

Innovation → creation of a **market for pollution rights**

Let's imagine that x must purchase from y rights to pollute the river they share

→ x' 's choice to purchase these rights is identical to its output choice (it cannot produce without them)

Net revenue for x per unit will be: $(P_x - r)$, where r is the payment the firm must make for each unit produced

17. Solutions to the externality problem: **Pollution rights**

Firm y must decide how many rights to sell to firm x, and will be paid r for each right:

$$\pi_y = p_y g(x_i, x_o) + r x_o$$

When y maximizes profits, it must also “choose” x output! First order condition for a maximum is:

$$\partial \pi_y / \partial x_o = p_y \partial g / \partial x_o + r = 0 \rightarrow r = -p_y \partial g / \partial x_o$$

The equilibrium solution is identical to that for the Pigouvian tax

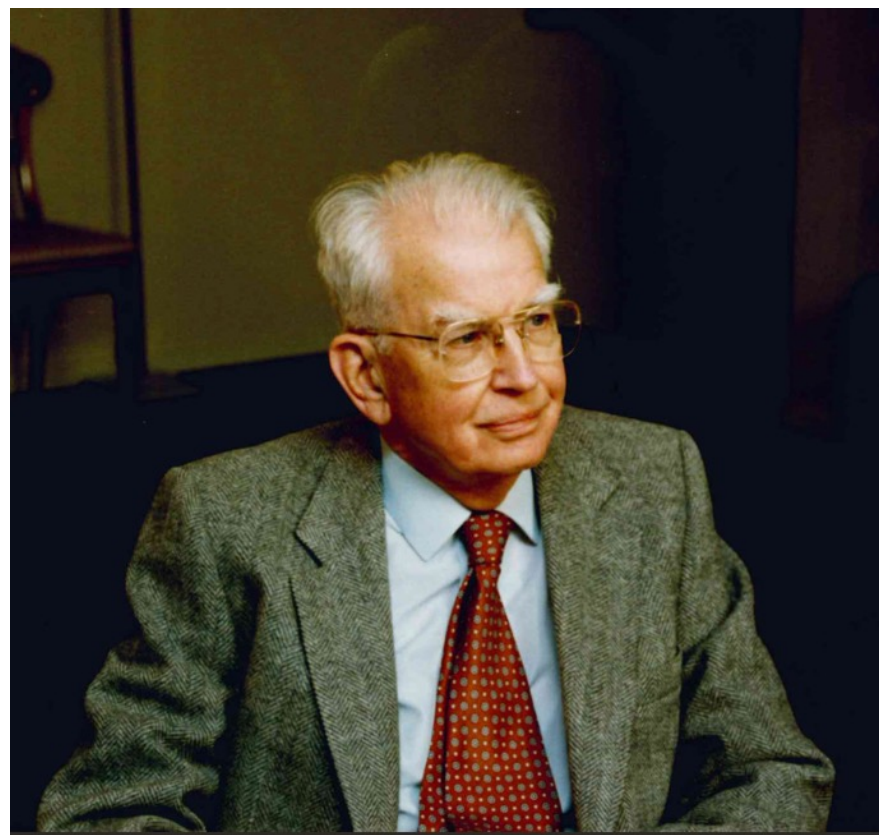
from firm x' 's point of view, it makes no difference whether it pays the fee to the government or to firm y

17. Solutions to the externality problem: **Pollution rights**

The Coase Theorem → Independence of initial rights

In 1960 R. Coase showed that the key feature of the pollution rights equilibrium is that these rights be well-defined and tradable with zero transaction costs.

The initial assignment is irrelevant
→ subsequent trading will yield the same efficient equilibrium.



Ronald H. Coase, 1991. Photo: Boo Jonsson

17. Solutions to the externality problem: **Pollution rights**

The Coase Theorem → Independence of initial rights

Let's prove this → Suppose that firm x is initially given x^T rights to produce (and to pollute), it can choose to use these for its own production, or it may sell some to firm y .

Now, profits for firm x are given by

$$\pi_x = p_x x_o + r(x^T - x_o) = (p_x - r)x_o + rx^T$$

$$\pi_x = (p_x - r)f(y_i) + rx^T$$

Profits for firm y will be

$$\pi_y = p_y g(x_i, x_o) - r(x^T - x_o)$$

17. Solutions to the externality problem: **Pollution rights**

The Coase Theorem → Independence of initial rights

Although this result may seem counterintuitive, it is the assertion that all mutually beneficial transactions will be completed in the absence of impediments to making bargains.

When transaction costs are high or when information is asymmetric, initial rights assignments will matter because the sorts of trading implied by the Coase theorem may not occur.

Individual Activity

In a piece of paper write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.